Problem 1: Calculate the equilibrium conversion and concentrations for each of the following reactions

(a) The liquid-phase reaction A +B ↔ C with $C_{A0} = C_{B0} = 2M$ and $K_c = 10 L/mol$.

(b) The gas-phase reaction A ↔ 3C carried out in a flow reactor with no pressure drop. Pure A enters at a temperature of 400K and 10 atm. At this temperature, $K_C = 0.25 (L/mol)^2$

Problem 2:

The gaseous reaction A → B has a unimolecular reaction rate constant of 0.0015 min^{-1} at 80^0F. This reaction is to be carried out in parallel tubes 10 ft long and 1 in inside diameter under a pressure of 132 psig at 260^0F. A production rate of 1000 lb/h of B is required. Assuming an activation energy of 25,000 cal/mol, how many tubes are needed if the conversion of A is to be 90%? Assume perfect gas law. A and B each have molecular weight of 58.